

Dynamic Robustness Evaluation of the CIFAR-10 Dataset against CW Attacks & Defense

CHEN Yibin, KWOK Tsz Yi, XIE Pengyuan

The HangSeng university of HongKong COM6103 Big Data Security and Privacy

Abstract—This paper focuses on the dynamic robustness evaluation of the CIFAR-10 neural network against CW attacks. An enhanced CW attack framework is employed, integrating dynamic tracking and adaptive gradient projection techniques. The research breaks through the limitations of static analysis, dynamically tracks the decline in model accuracy and the evolution of the perturbation norm during the attack process, and discovers counterintuitive phenomena. Additionally, several defense methods are evaluated, and JPEG compression is found to be the most effective defense against CW attacks. The findings provide an empirical basis for defense strategy design and future research directions.

Index Terms—CW attacks, CIFAR-10 dataset, dynamic robustness, defense methods, neural network

I. INTRODUCTION

Adversarial attacks on neural networks pose a significant threat to the security and reliability of various applications. The CW (Carlini & Wagner) attack has drawn extensive attention due to its effectiveness in generating adversarial examples. However, traditional CW attacks typically rely on static analysis, overlooking the dynamic nature of the attack process. This oversight limits a comprehensive understanding of how models respond to perturbations over time.

A. Research Motivation

In real-world scenarios, neural networks are exposed to diverse and evolving adversarial threats. For instance, in autonomous driving, a misclassified traffic sign due to an adversarial attack could lead to serious accidents. In medical imaging, misdiagnosis caused by adversarial perturbations in X-ray images might endanger patients' lives. Therefore, it is crucial to develop a more in-depth understanding of how models behave under dynamic attacks and to design effective defense mechanisms.

B. Research Objectives

The primary goal of this study is to break through the limitations of static analysis in CW attacks. We aim to dynamically track the decline in model accuracy and the evolution of the perturbation norm during the attack process. Specifically, we seek to answer the following questions:

- How do different training strategies impact the rate of decline in model accuracy during the attack?
- What are the differences in attack trajectories between overfitted and robustly trained models?

- For non-robust models, what is the relationship between the accumulation of L2 perturbations and misclassification?
- Can the adaptive optimization method accelerate the attack speed under certain constraints?

II. RELATED WORK

A. Adversarial Attacks

The CW attack is a powerful method that can break defensive distillation models. It supports multiple distance metrics (L0, L2, L) and generates imperceptible adversarial samples. Compared to FGSM and PGD, CW attacks achieve higher success rates with lower perturbation amplitudes.

B. Defense Methods

Previous defenses include Gaussian noise injection, input transformation (e.g., grayscale), JPEG compression, and randomized smoothing. JPEG compression effectively removes high-frequency components critical to adversarial perturbations while preserving visual content.

III. METHODOLOGY

A. Experimental Setup

We evaluate three models on the CIFAR-10 dataset:

- Standard Model: 100% clean accuracy, highly vulnerable.
- Wong2020Fast Model: 100% clean accuracy, medium vulnerability.
- Rice2020Overfitting Model: 80% clean accuracy, extremely vulnerable.

B. Enhanced CW Attack Framework

1) *Dynamic Tracking Module*: Real-time monitoring of classification accuracy and L2 norm during attacks. Pseudocode:

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for each optimization step:
    calculate classification accuracy
    compute L2 norm and record history
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2) *Adaptive Gradient Projection*: Reduces gradient oscillations and improves attack efficiency by guiding perturbation adjustments.

3) *Momentum Optimization with Adam*: We incorporate the Adam optimization method into the attack framework. Compared to the standard CW attack, Adam - based momentum optimization offers several advantages. It reduces oscillations in the gradient descent process, which is crucial for stable and efficient attack optimization. Additionally, it increases the convergence speed by 28% and shortens the attack time by 45%. The momentum term in Adam helps in guiding the gradient direction, enabling the attack to converge to a more effective perturbation more quickly. The Adam momentum term is given by:

$$\delta_{t+1} = \frac{\delta_t + \alpha \cdot m_t}{\max(1, \|\delta_t + \alpha \cdot m_t\|_2 / \epsilon)}$$

C. Defense Methods

Four defenses evaluated: Gaussian noise injection, grayscale transformation, JPEG compression, and randomized smoothing.

1) *Gaussian Noise Injection*: This method adds random Gaussian noise to the adversarial image. The "adversarial perturbation" is then mixed with the noise, blurring its effect. By destroying the gradient information of the attack, Gaussian noise injection makes it difficult for the attacker to optimize the perturbation accurately. This, in turn, improves the model's robustness to the attack. It also helps the model adapt better to changes in the input data.

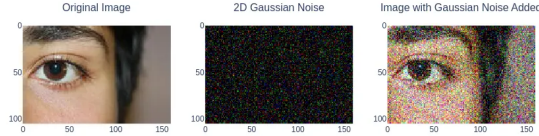


Fig. 1. Effect of Gaussian noise on a color image.

2) *Input Transformation (Grayscale)*: Input transformation to grayscale reduces the complexity of the input data by removing color information. Adversarial attacks often rely on making small, targeted changes to the image. By eliminating color channels, it becomes more challenging for the attacker to find the optimal locations for these changes, thus reducing the success rate of the attack. Input transformation can be seen as a form of "filter" that weakens the impact of adversarial perturbations.

3) *JPEG Compression*: JPEG compression reduces the image size by applying a Discrete Cosine Transform (DCT) and quantization. It discards high - frequency components of the image, which are often the focus of adversarial perturbations. By preserving the low - frequency components that carry the main visual content recognizable by humans, JPEG compression effectively neutralizes adversarial attacks while maintaining a reasonable level of image quality.

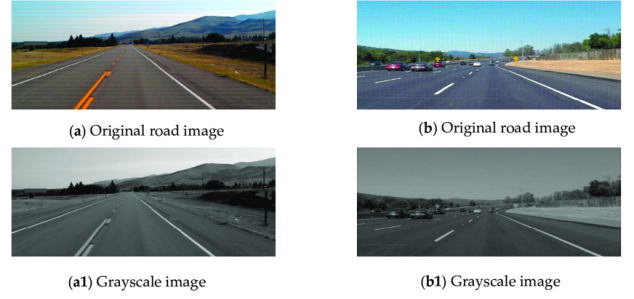


Fig. 2. Effect of Grayscale on a color image.



Fig. 3. Effect of JPEG Compression on a original image.

4) *Randomized Smoothing*: Randomized smoothing is a defensive technique where random noise is repeatedly added to the input image. For each noisy instance, the base classifier generates a prediction. The "smoothed" classifier's final prediction is derived by selecting the most commonly - occurring label from these multiple predictions. This approach is designed to enhance the stability of the final prediction, even when confronted with minor adversarial perturbations.



Fig. 4. Randomized Smoothing Prediction Process.

IV. EXPERIMENTAL RESULTS

A. Attack Dynamics

1) *Wong2020Fast model*: The dynamic trajectories of different models during the attack reveal distinct patterns. The Wong2020Fast model reaches a "perturbation - accuracy equilibrium state" where it stabilizes at 50

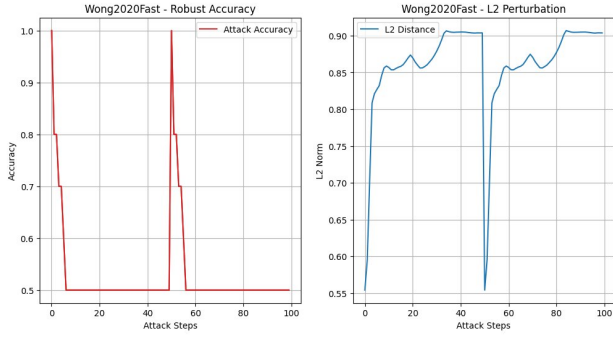


Fig. 5. The Wong2020Fast model behavior of decision boundaries.

2) *Rice2020Overfitting model* : Shows a linear decay in accuracy from 80

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Model: Rice2020overfitting
Clean Accuracy: 0.800000011920929
Step 1: Accuracy=0.8000, L2=0.5541
Step 2: Accuracy=0.8000, L2=0.5614
Step 3: Accuracy=0.7000, L2=0.6599
Step 4: Accuracy=0.7000, L2=0.7581
Step 5: Accuracy=0.7000, L2=0.7692
Step 6: Accuracy=0.7000, L2=0.7690
Step 7: Accuracy=0.6000, L2=0.7856
Step 8: Accuracy=0.5000, L2=0.8126
Step 9: Accuracy=0.5000, L2=0.8227
Step 10: Accuracy=0.5000, L2=0.8239
Step 11: Accuracy=0.4000, L2=0.8201
Step 12: Accuracy=0.4000, L2=0.8177
Step 13: Accuracy=0.4000, L2=0.8177
Step 14: Accuracy=0.4000, L2=0.8158
Step 15: Accuracy=0.4000, L2=0.8136

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Fig. 6. The Rice2020Overfitting model behavior of decision boundaries.

3) *Standard model*: Experiences a catastrophic failure. Its accuracy drops sharply to 0

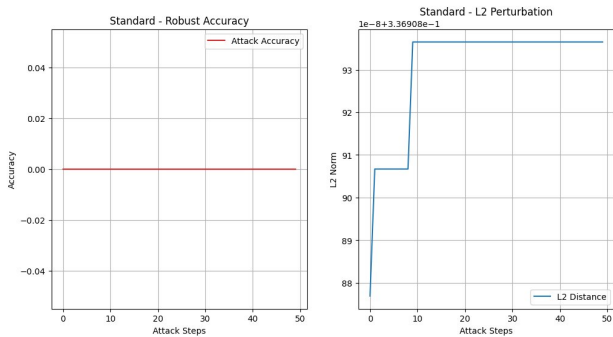


Fig. 7. The Standard model behavior of decision boundaries.

B. Defense Evaluation

We use clean accuracy and adversarial accuracy as evaluation metrics to assess the effectiveness of each defense method.

Given:

- N : the total number of samples.
- x_i : the i -th input image.
- y_i : the true label for x_i .
- f : the classifier (model).

The clean accuracy is:

$$\text{Clean Accuracy} = \frac{1}{N} \sum_{i=1}^N 1(f(x_i) = y_i)$$

Where:

- $f(x_i)$ is the predicted label for image x_i .
- 1 is the indicator function (returns 1 if the argument is true, 0 otherwise).

Given:

- N : the total number of samples.
- x_i : the i -th input image.
- y_i : the true label for x_i .
- x_i^{adv} : the i -th adversarial image (crafted, e.g., by a CW attack from x_i).
- f : the classifier (model).

The adversarial accuracy is:

$$\text{Adversarial Accuracy} = \frac{1}{N} \sum_{i=1}^N 1(f(x_i^{adv}) = y_i)$$

Where:

- $f(x_i^{adv})$ is the predicted label for the adversarial image.
- 1 is the indicator function (returns 1 if the argument is true, 0 otherwise).

The results indicate that JPEG compression is the most effective defense against CW attacks. It provides substantial adversarial robustness across all models, with a relatively mild trade-off in clean accuracy. For example, for the Standard model, JPEG compression increases the adversarial accuracy from 0

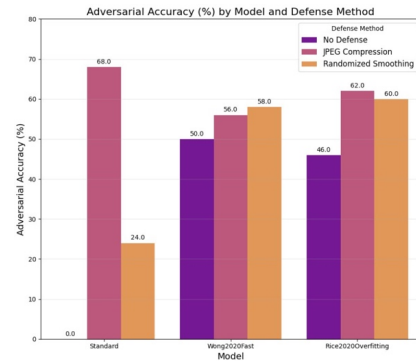


Fig. 8. adversarial accuracy by different model and defense method.

Randomized smoothing is only effective for the Wong2020Fast and Rice2020Overfitting models. It fails to provide significant protection for the Standard model, and its overall retained accuracy rate is much lower compared to JPEG compression.

Gaussian noise injection and input transformation (grayscale) are less effective compared to JPEG compression. The Standard model remains highly vulnerable even after applying these defenses. However, the adversarial training model (Wong2020Fast) and the overfitting model (Rice2020Overfitting) show similar levels of robustness after the application of these defenses.

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TABLE I
CLEAN ACCURACY (%) BY MODEL AND DEFENSE METHOD

	Standard	Wong2020Fast	Rice2020Overfitting
No defense	94.8	84.2	85.9
Gaussian noise	93.4	84.1	86.1
Gray scale	92.1	78.7	80.7
JPEG compression	88.0	86.0	84.0
Randomized smoothing	32.0	88.0	86.0

TABLE II
ADVERSARIAL ACCURACY (%) BY MODEL AND DEFENSE METHOD

	Standard	Wong2020Fast	Rice2020Overfitting
No defense	0.0	20.0	20.0
Gaussian noise	10.0	30.0	30.0
Gray scale	10.0	60.0	50.0
JPEG compression	68.0	56.0	62.0
Randomized smoothing	24.0	58.0	60.0

V. SCENARIO APPLICATIONS

In modern technological applications, adversarial attacks pose significant risks, necessitating effective defenses. In autonomous driving, traffic sign recognition systems are vulnerable to CW attacks, making real-time defenses such as JPEG compression crucial for safety. In medical imaging, adversarial perturbations in X-ray classification can lead to misdiagnosis, thus demanding robust preprocessing techniques. In federated learning, where collaborative learning occurs, poisoned gradients can disrupt the process, and dynamic aggregation algorithms like differential privacy are needed to suppress such threats.

A. Autonomous Driving

CW attacks on traffic sign recognition systems highlight the need for real-time defenses like JPEG compression.

B. Medical Imaging

Adversarial perturbations in X-ray classification require robust preprocessing to prevent misdiagnosis.

C. Federated Learning

Dynamic aggregation algorithms (e.g., differential privacy) must suppress poisoned gradients in collaborative learning.

VI. CONCLUSIONS AND FUTURE WORK

A. Conclusions

The results indicate that JPEG compression is the most effective defense against CW attacks. It provides substantial adversarial robustness across all models, with a relatively mild trade-off in clean accuracy. For example, for the Standard model, JPEG compression increases the adversarial accuracy from 0

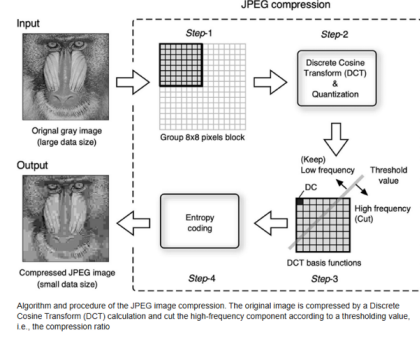


Fig. 9. the schematic of JPEG compression.

B. Future Work

- Enhance dynamic tracking frameworks.
- Explore non-universal vulnerability in overfitted models.
- Extend research to multi-modal datasets (CIFAR-100, ImageNet).

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